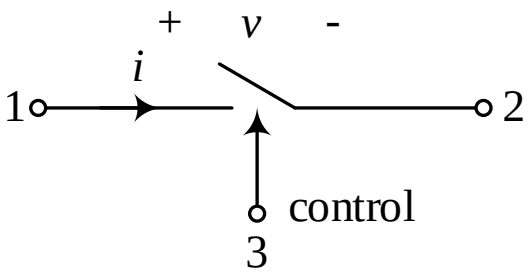
Chapter 2

Power Semiconductor Switches

Instructor: Le Hoa Nguyen, Ph.D.

Ideal switch

- In power electronic systems, power semiconductor devices are used as switches (i.e., it is either ON or OFF)
- An ideal switch has following properties



1) ON sate (the switch is closed)

- $i_{ON} = \infty$
- $v_{ON} = 0$
- $R_{ON} = 0$

2) OFF sate (the switch is opened)

- $i_{OFF} = 0$
- $v_{OFF} = \infty$

3) Dynamic characteristics

- $t_{ON} = 0$
- $t_{OFF} = 0$
- $dv/dt = \infty$
- $di/dt = \infty$

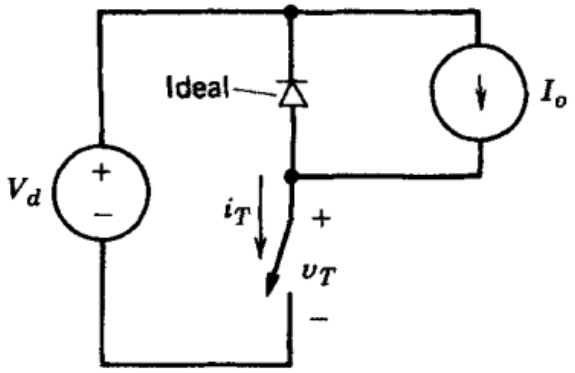
4) Able to close or open switch by control signal with

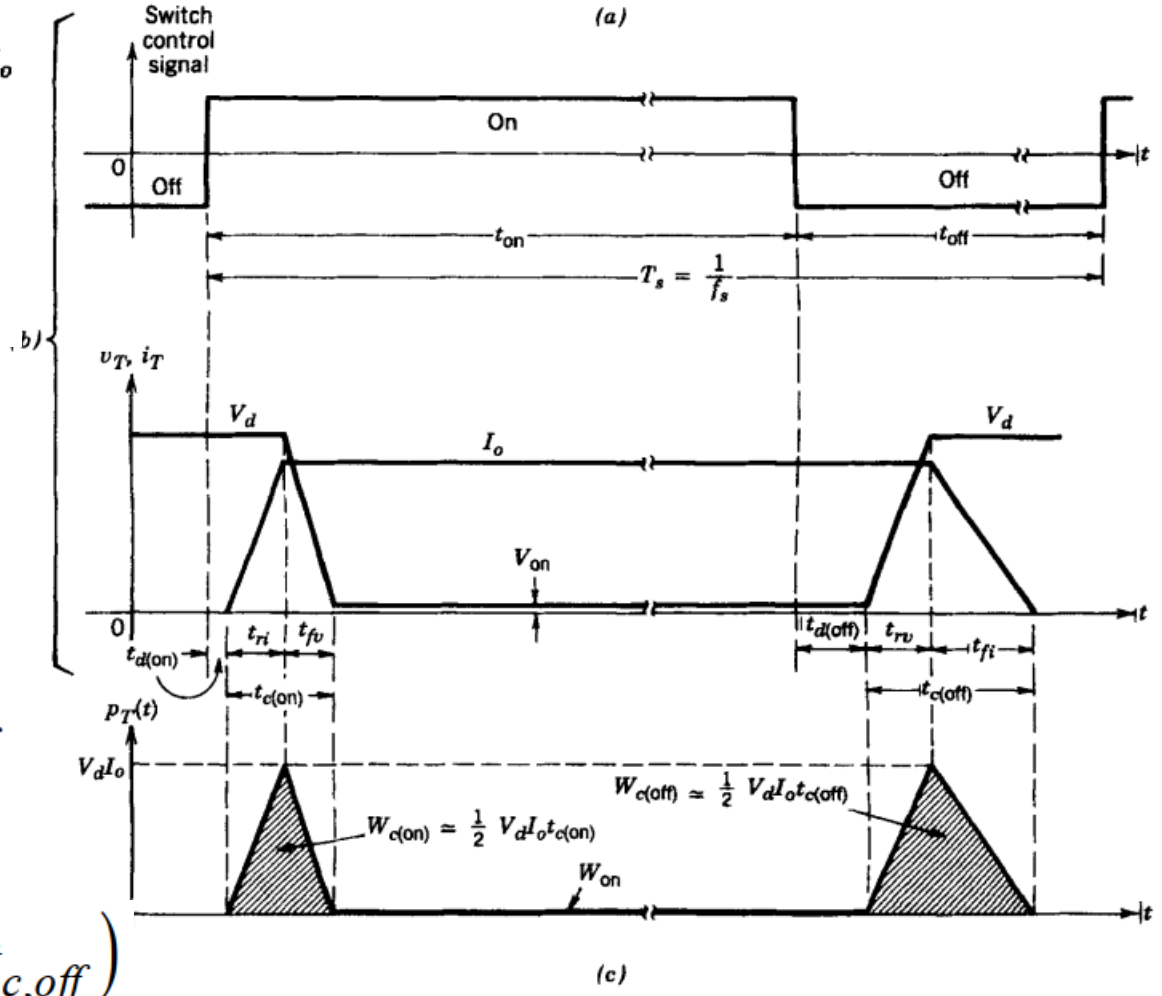
- $i_G = 0$
- $v_G = 0$

5) Loss power during ON state, OFF state and switching is zero

Read carefully Section 2.4 of textbook for more detail explaination

Power dissipation in a real semiconductor device





- Turn-on energy

$$W_{c,on} = \frac{1}{2} V_d I_o t_{c,on}$$

- Turn-off energy

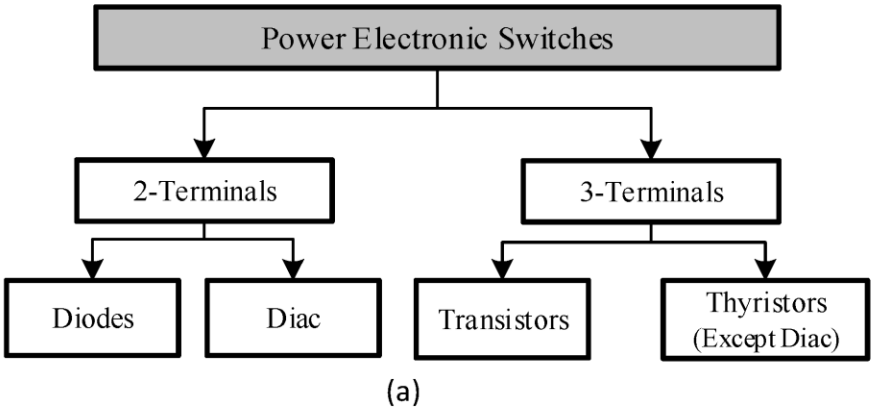
$$W_{c,off} = \frac{1}{2} V_d I_o t_{c,off}$$

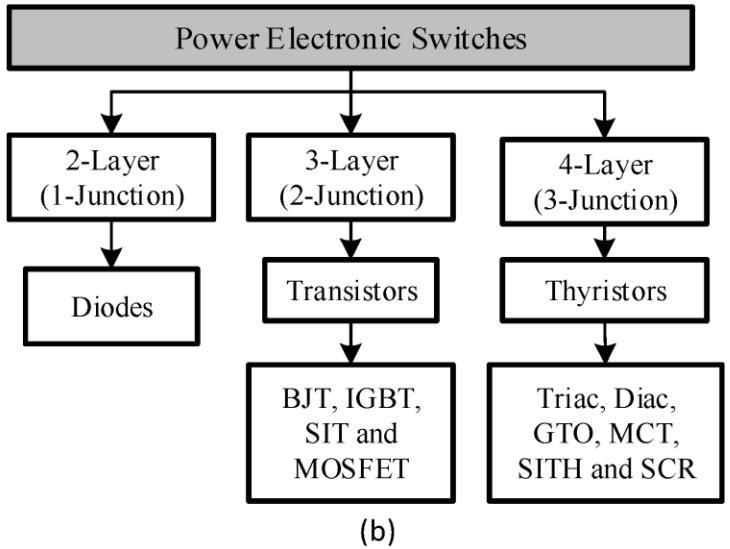
- Switching loss power

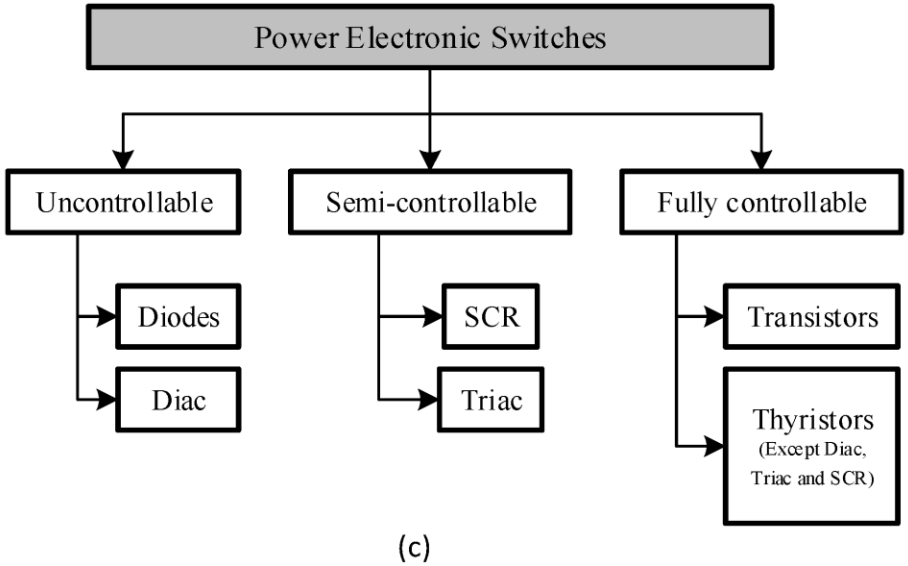
$$P_s = \frac{1}{2} V_d I_o f_s \left(t_{c,on} + t_{c,off} \right)$$

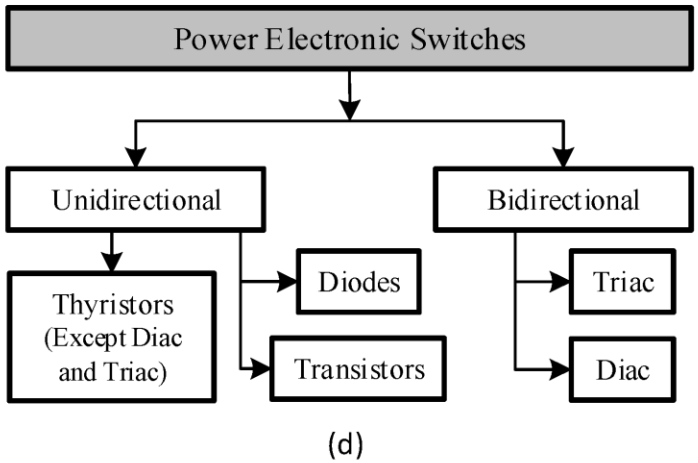
- Conduction loss power $P_{on} = V_{on} I_o \frac{t_{on}}{T_s}$

Classficiation of power switches



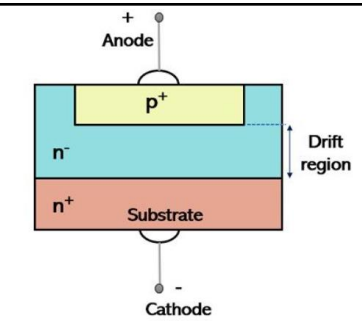






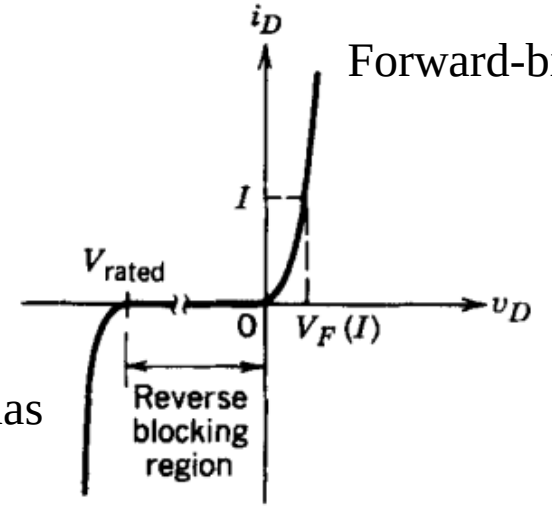
Source: Jiya and Gouws, "*Overview of power electronic switches: A summary of the past, state-of-art and illumination of the future*", Micromachines, 2020

Diodes

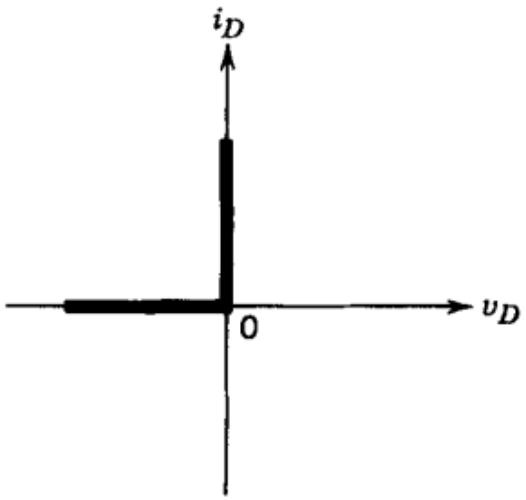


(a)

Reverse-bias



(b)

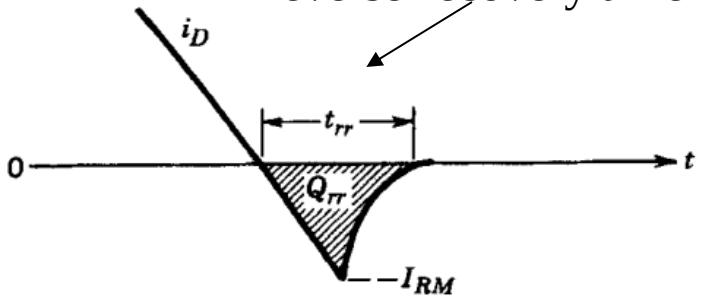


(c)

Figure 2-1 Diode: (*a*) symbol, (*b*) *i*–*v* characteristic, (*c*) idealized characteristic.



Reverse-recovery time



Diode turn off

Diodes

Types of diodes

Line-frequency diodes:

- On state voltage: very low (below 1V)
- Large t_{rr} (about 25 μs) (very low response)
- Very high current ratings (up to 6kV)
- Very high voltate ratings (up to 8kV)
- Acceptale for line-frequency applications (e.g., rectifiers)

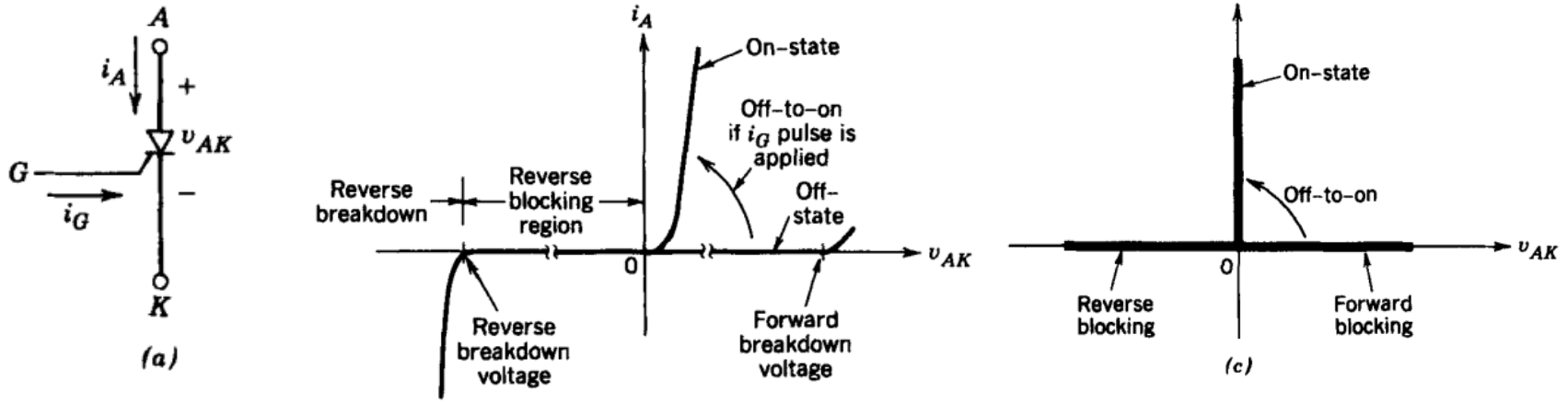
Fast recovery diodes:

- Very low t_{rr} (below 1 μs)
- Nomarlly used in high frequency circuits
- Power levels at several hundred volts and several hundred amps

Schottky diodes:

- Very low forward voltage drop (typically 0.3 V)
- Limited blocking voltage (20-100V)
- Used in low voltage, high current applications

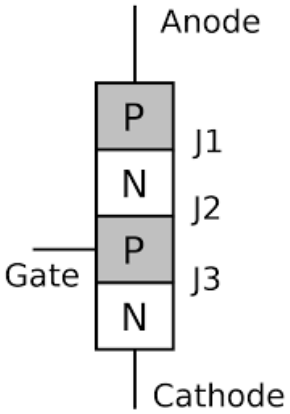
Thyristors



Thyristor: (*a*) symbol. (*b*) *i*–*v* characteristics, (*c*) idealized characteristics.

- Turn on by positive gate current (pulse)

  *Device must be in forward blocking state
- Turn off by anode current



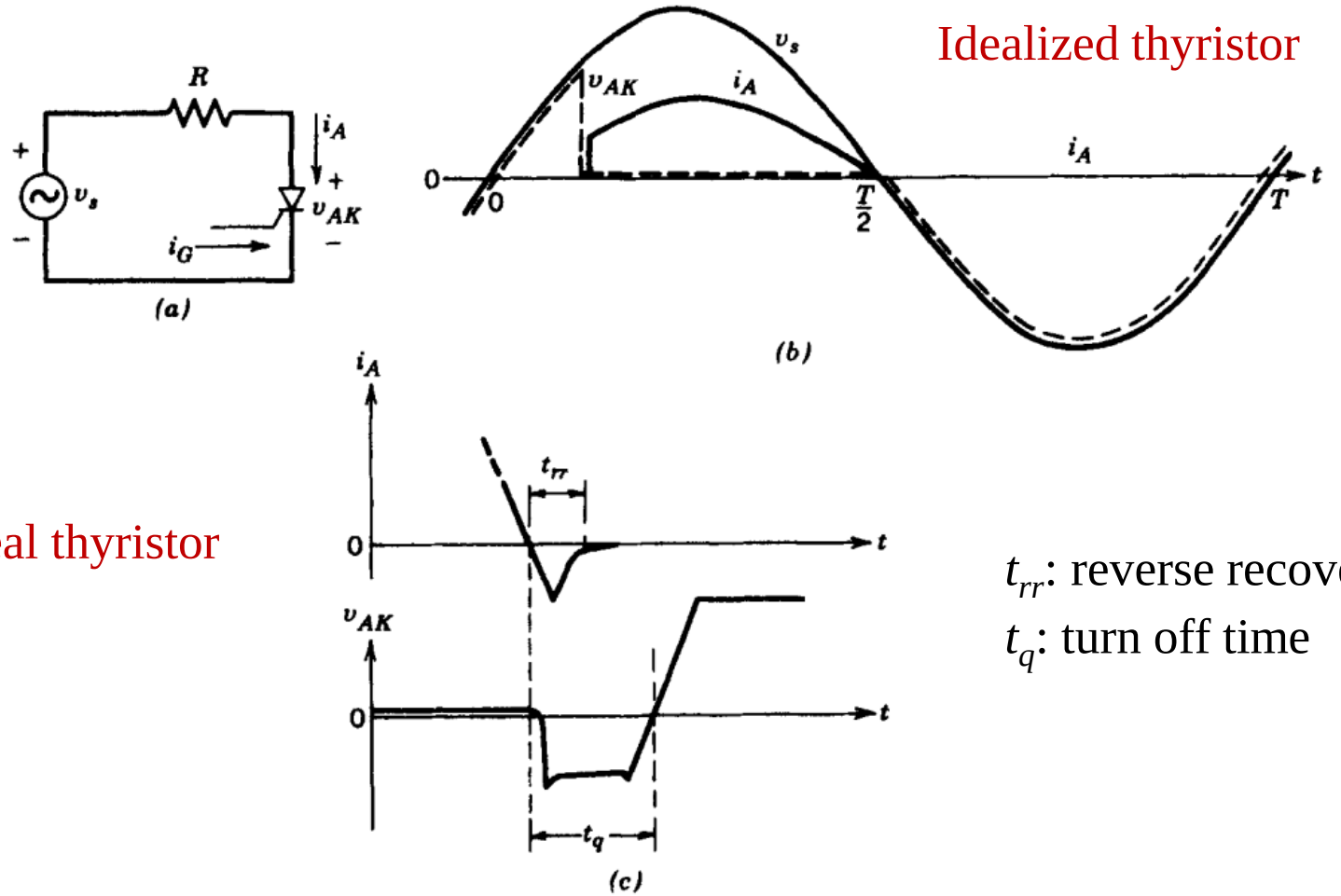






Thyristors

Idealized thyristor



Real thyristor

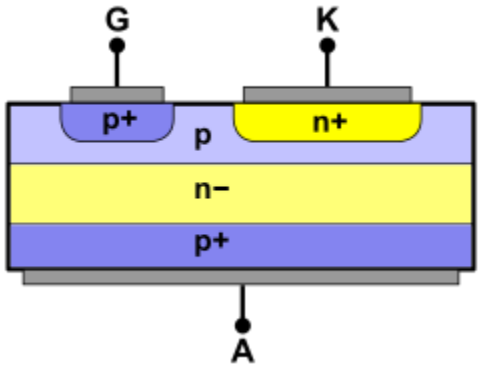
t_{rr}: reverse recovery time

t_q: turn off time

SWITCHING				
PARAMETER	SYMBOL	TEST CONDITIONS	VALUES	UNITS
Typical turn-on time	t_{gt}	T_J = 25 °C	0.9	µs
Typical reverse recovery time	t_{rr}	T_J = 125 °C	4	
Typical turn-off time	t_q		110	

Gate-Turn-Off (GTO) thyristors

- Turn on by a short-duration gate current pulse, once in the on-state, the GTO may stay on without any futher gate current
- Turn off by a large negative gate current for a few μs (i.e., $v_{GK} < 0$)
- GTO is a controllable switch (it can turn on and turn off through i_G)
- v_{ON} = 2-3V, switching time ~ 25μs, rating voltage up to 4.5 kV, rating current up to few kA



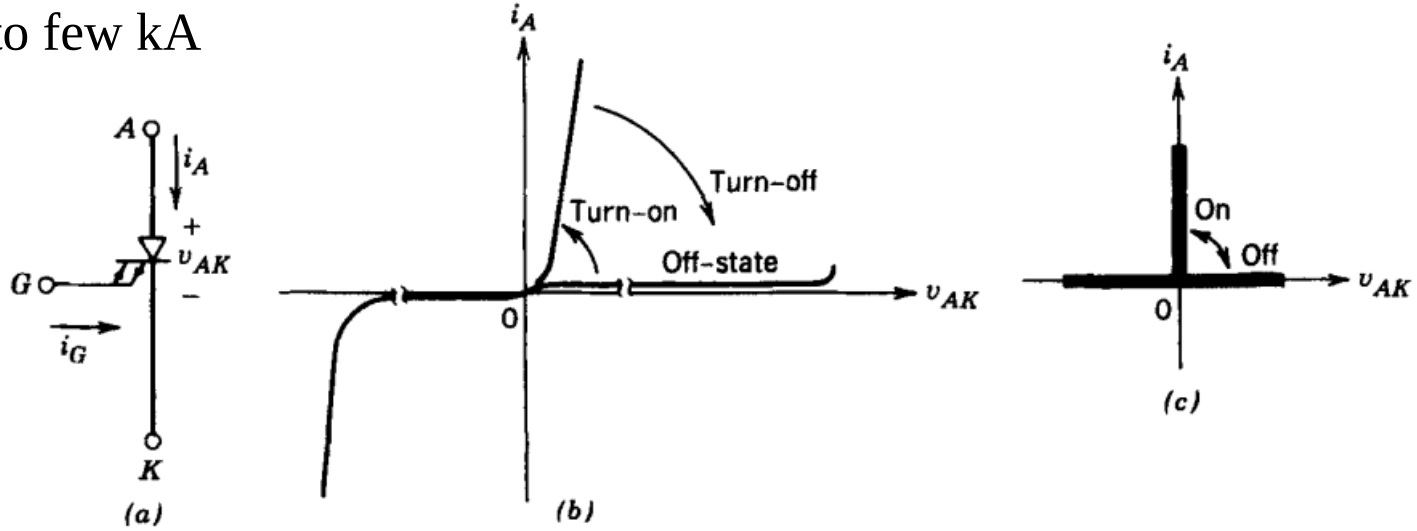
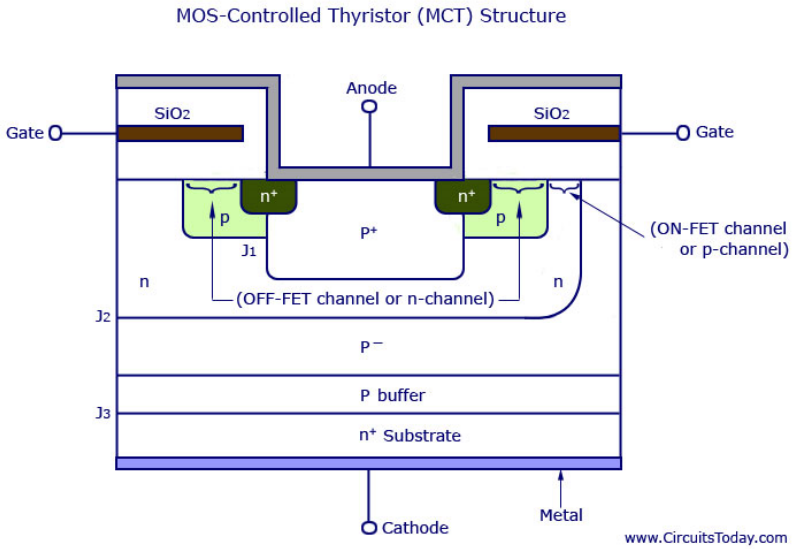


Figure 2-10 A GTO: (*a*) symbol, (*b*) *i–v* characteristics, (*c*) idealized characteristics.

MOS-Controlled Thyristors (MCT)

- MCT has many properties of a GTO
- MCT has two principle advances over the GTO, including much simple drive requirements (no large negative gate current required to turn off like the GTO) and faster switching speed.
- Small on-state voltage, switching time ~ few μs



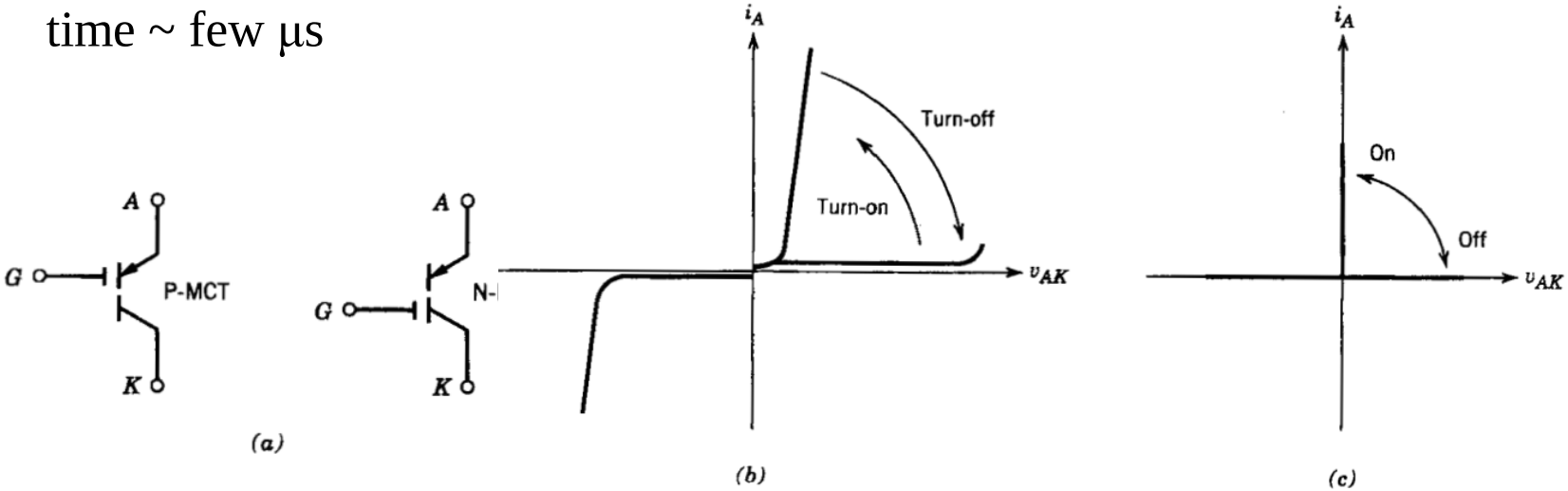
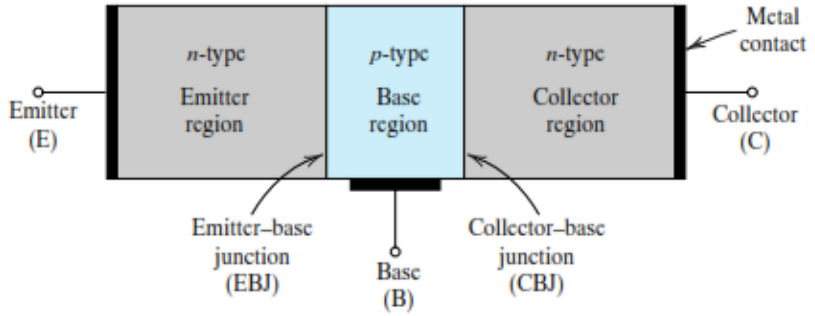
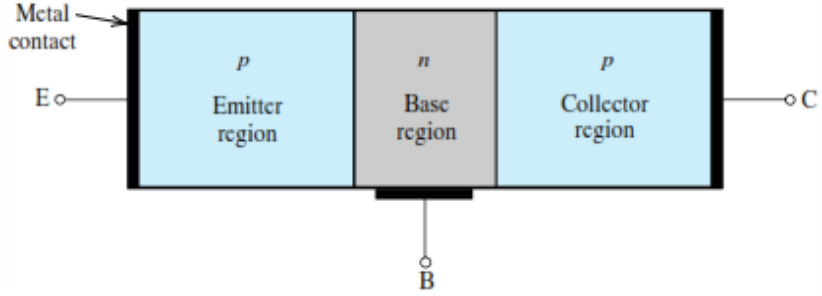


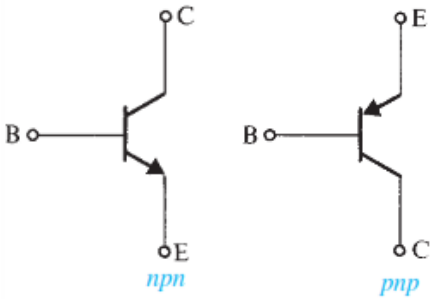
Figure 2-13 An MCT: (*a*) circuit symbols, (*b*) $i-v$ characteristic, (*c*) idealized characteristics.

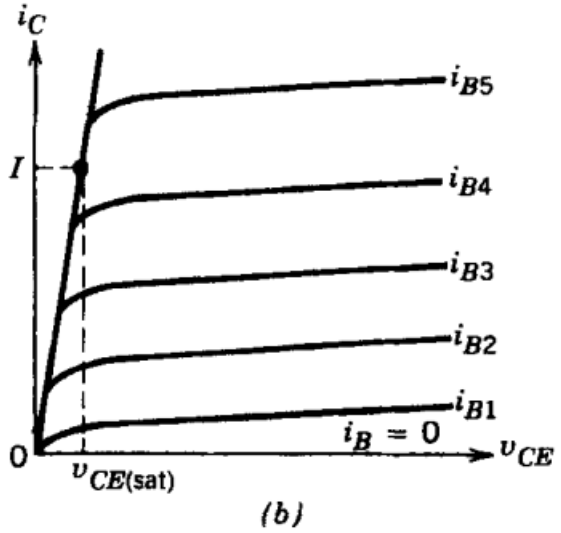
Bipolar Junction Transistor (BJT)

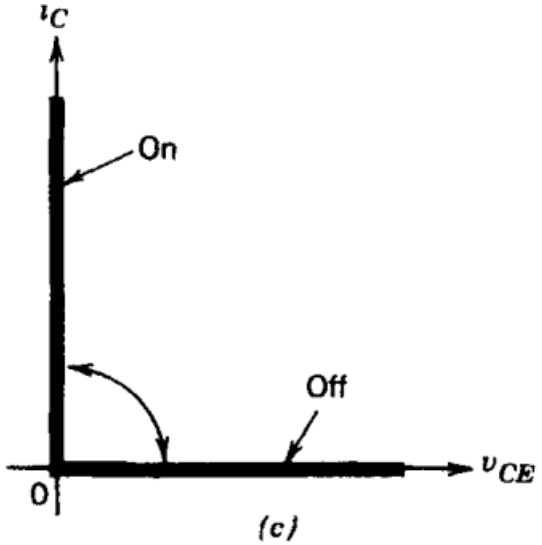




- BJT is a current-controlled devices
- On-state: A sufficient large base current ($i_B > i_C/h_{FE}$) is required to supply continuously, $v_{CE(\text{sat})}$ is usually 1-2V. The BJT operates in the staturation region
- Off-state: $i_B = 0$, the BJT operates in the cut-off region

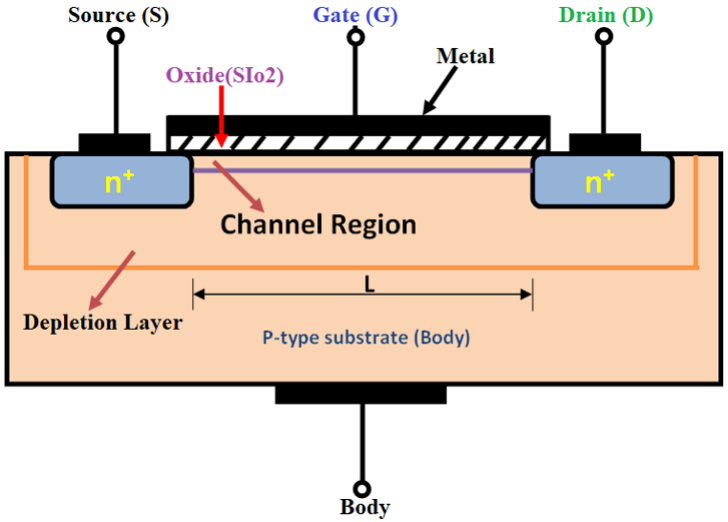






Metal-Oxide-Semiconductor Field Effect Transistor (MOSFET)

- MOSFET is a voltage-controlled device
- Control voltage v_{GS} is usually from 5 to 20V
- Fast, switching time in the range of few tens *ns* to a few hundred *ns*
- On-resistance increase rapidly with the device blocking voltage rating (r_{ON} ~ $V^{2.5\text{-}2.7}$) => conduction loss



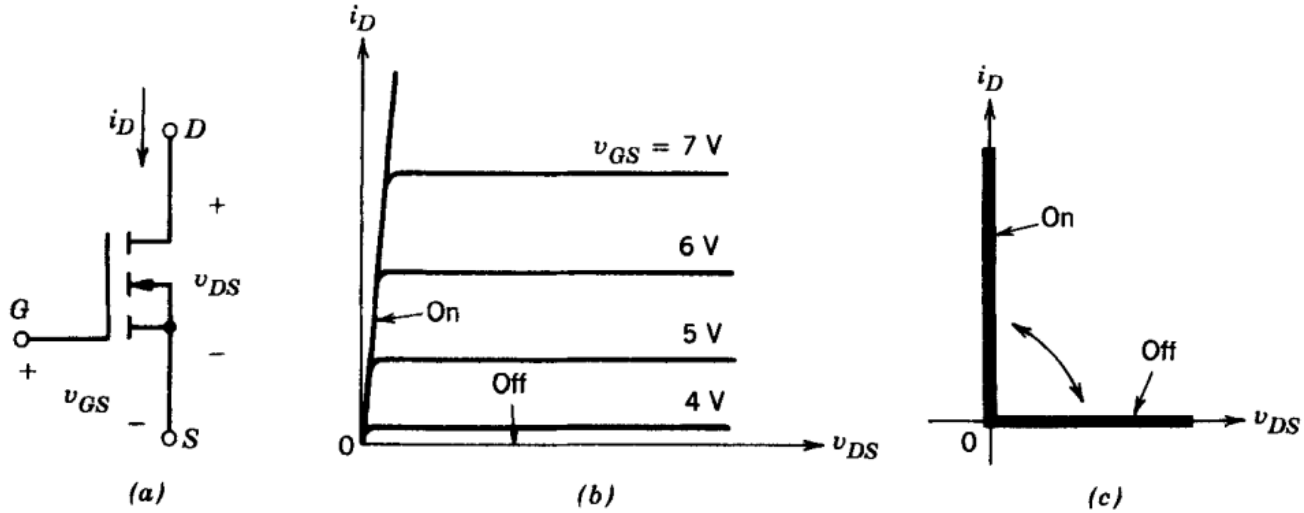
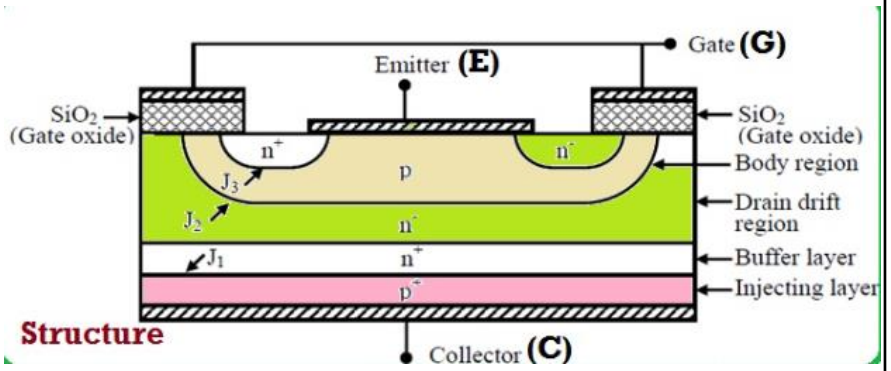
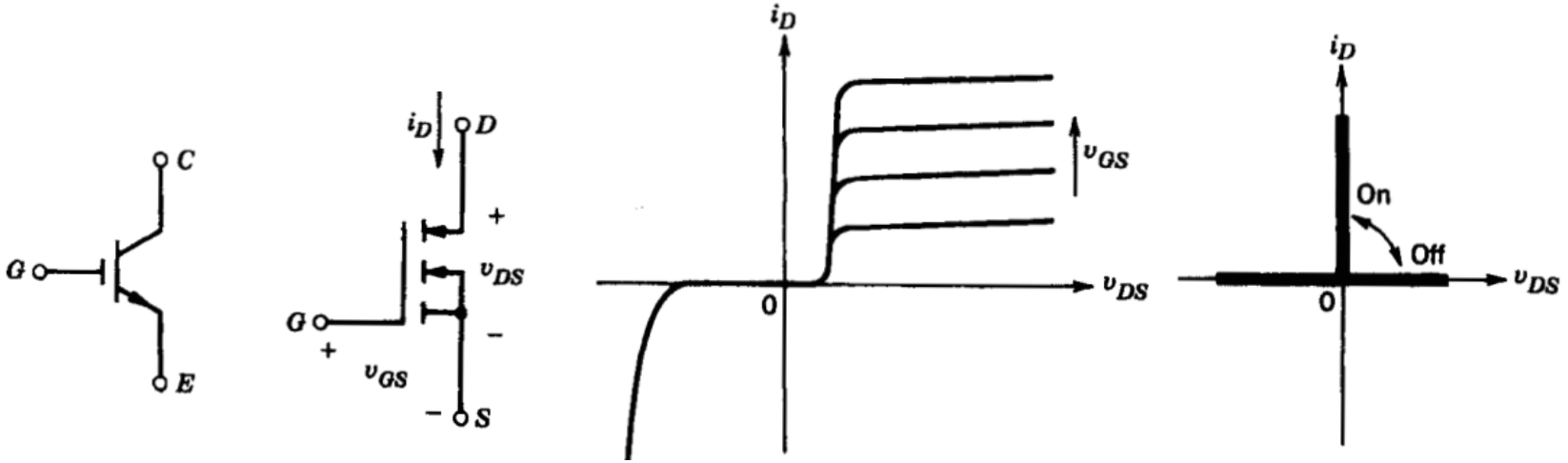


Figure 2-9 *N*-channel MOSFET: (*a*) symbol, (*b*) *i*–*v* characteristics, (*c*) idealized characteristics.

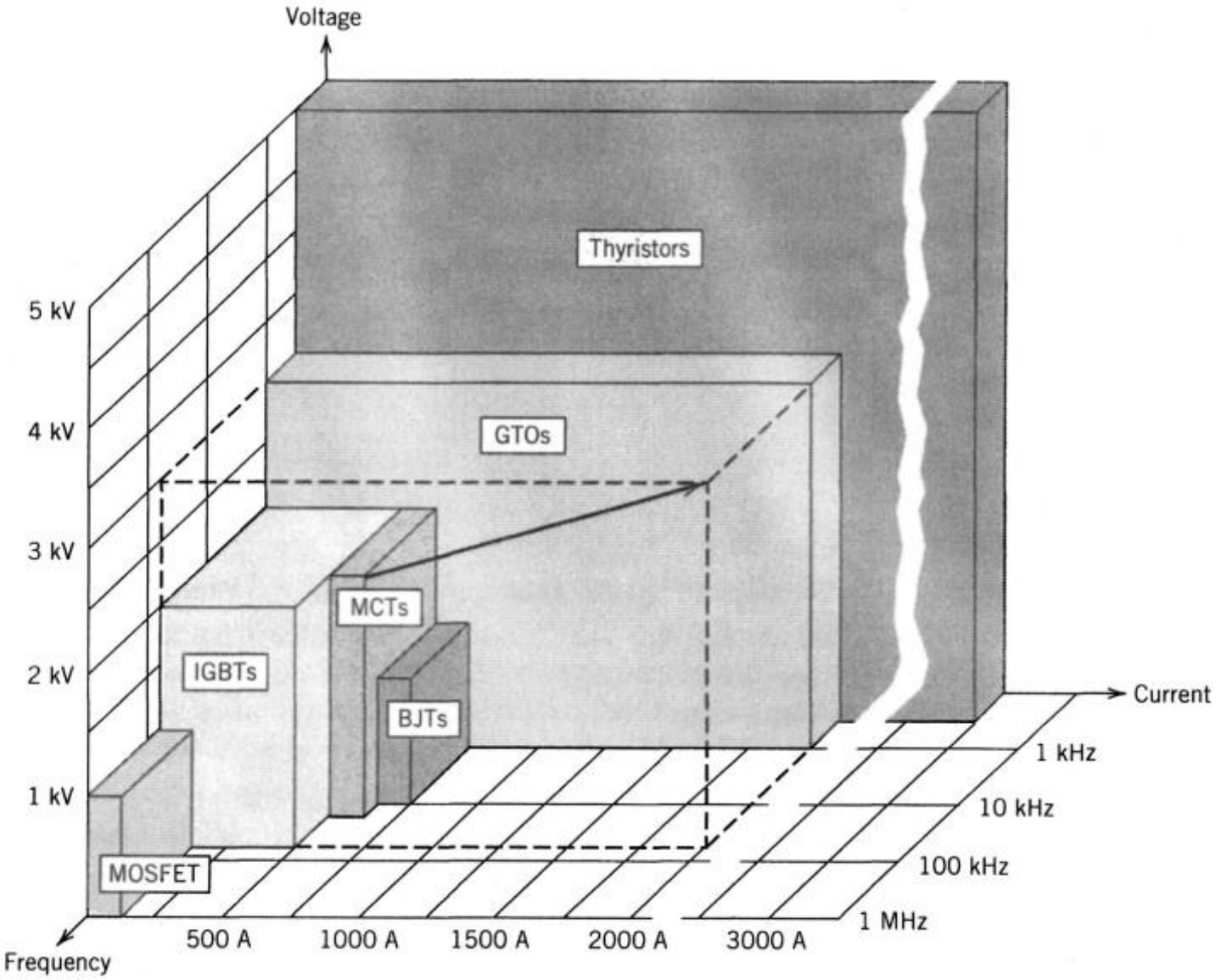
Insulated Gate Bipolar Transistor (IGBT)

- High impedance gate (like MOSFET) => require small amount energy to switch the device
- Small on-state voltage (like BJT) (e.g., v_{ON} = 2-3V in a 1000V device)
- Turn-on and turn-off time ~ 1 μs.

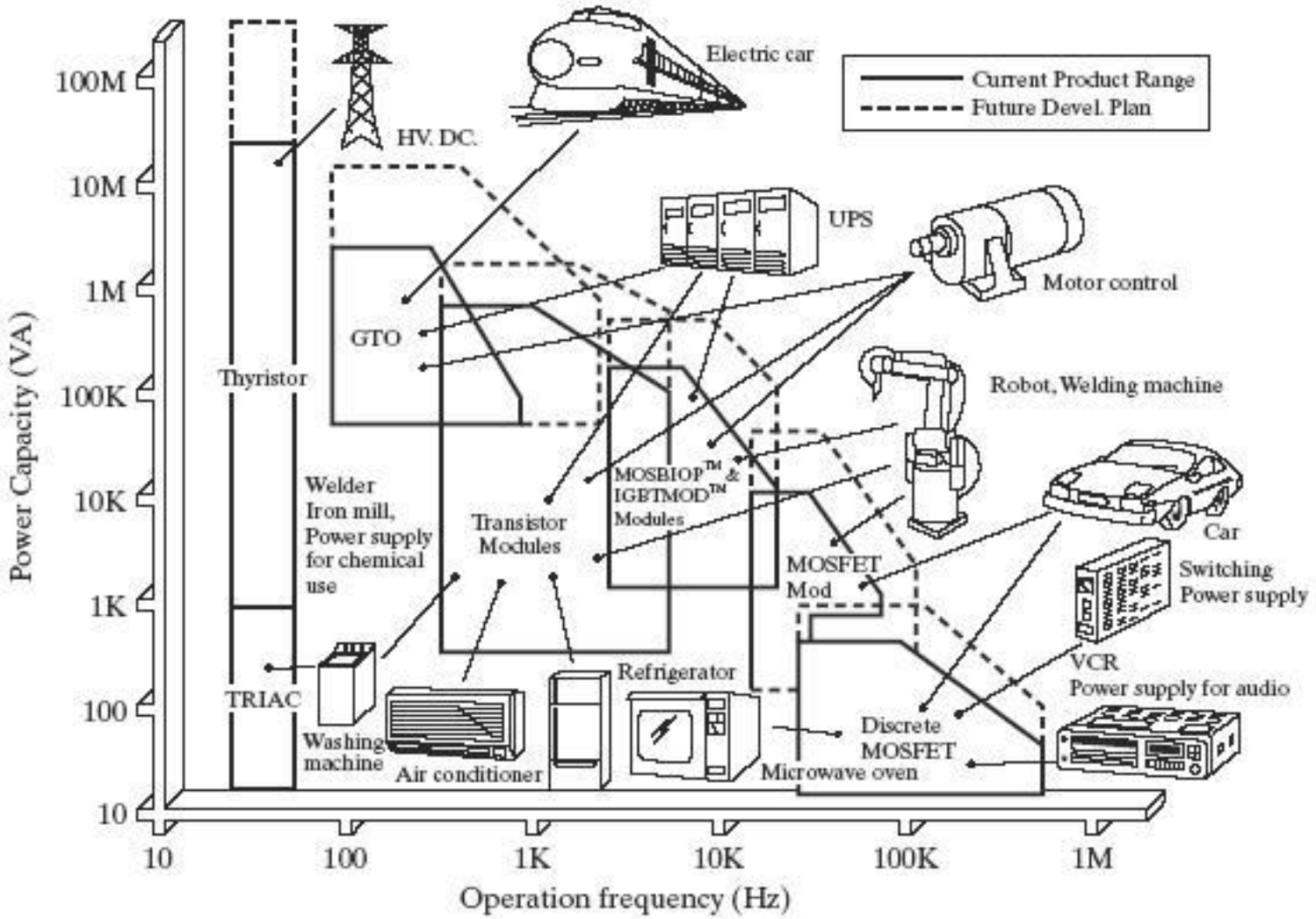




Comparison of controllable switches



Comparison of controllable switches



Assignment

Complete the following table

Parameter	Thyristor	GTO	MCT	BJT	MOSFET	IGBT
Operating freq.						
Trigger signal						
On-state voltage drop						
Voltage rating						
Current rating						
Switching speed						
Symbol						